THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 29	$(\alpha+\beta)^2-2\alpha\beta=7^2-2(10)$
2	**(d)** $\frac{5}{6}$	$\frac{\alpha+\beta}{\alpha\beta}$
3	**(c)** 10	$2^2-7(2)+k=0$
4	**(a)** 3	$2^2+2k-10=0$
5	**(c)** 14	Remainder $=p(2)=14$
6	**(d)** 4	Other zero $=7-3$
7	**(c)** 25	$(\alpha+\beta)^2-2\alpha\beta=7^2-2(12)$
8	**(a)** $\frac{5}{4}$	$\frac{\alpha+\beta}{\alpha\beta}$
9	**(b)** 15	$3^2-8(3)+k=0$
10	**(d)** -1	$3^2+3k-6=0$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\alpha+\beta=8$, $\alpha\beta=12$ (1); $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=40$ (1).	1+1
12	$p(3)=0$: $-15+k=0$ (1) so $k=15$ (1).	1+1
13	New sum $=2(3)=6$, product $=4(2)=8$ (1); polynomial x^2-6x+8 (1).	1+1
14	$\alpha+\beta=10$, $\alpha\beta=21$ (1); $\alpha^2+\beta^2=(\alpha+\beta)^2-2\alpha\beta=58$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\alpha+\beta=6$, $\alpha\beta=8$ (1); $\frac{\alpha+\beta}{\alpha\beta}=\frac{3}{4}$ (1); $\alpha^2+\beta^2=20$ (1).	1+1+1
16	Sum of new zeroes $=\frac{\alpha+\beta}{\alpha\beta}=\frac{5}{6}$ (1); product $=\frac{1}{\alpha\beta}=\frac{1}{6}$ (1); polynomial $x^2-(\frac{5}{6})x+(\frac{1}{6})$ (1).	1+1+1
17	$p(6)=0$: $36-66+30=0$ gives $k=-11$ (1); sum of zeroes $=-k=11$ (1); other zero $=5$ (1).	1+1+1
	OR: $\alpha+\beta=6$, $\alpha\beta=5$ (1); $\frac{\alpha+\beta}{\alpha\beta}=\frac{6}{5}$ (1); $\alpha^2+\beta^2=26$ (1).	1+1+1
18	Sum of new zeroes $=\frac{\alpha+\beta}{\alpha\beta}=\frac{7}{6}$ (1); product $=\frac{1}{\alpha\beta}=\frac{1}{6}$ (1); polynomial $x^2-(\frac{7}{6})x+(\frac{1}{6})$ (1).	1+1+1
	OR: $p(-5)=0$: $25-15-10=0$ gives $k=3$ (1); sum of zeroes $=-k=-3$ (1); other zero $=2$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Quadratic factor $(x-1)(x-2)=x^2-3x+2$ (1); divide to get $x-3$ (2); third zero 3 (1); zeroes $1, 2, 3$ (1).	1+2+1+1
	OR: Sum $=5$, product $=k$ (1); $k=2\times5=10$ (2); polynomial $x^2-5x+10$ (2).	1+2+2
20	Quadratic factor $(x+1)(x-2)=x^2-x-2$ (1); divide to get $x-5$ (2); third zero 5 (1); zeroes $-1, 2, 5$ (1).	1+2+1+1
	OR: Sum $=6$, product $=k$ (1); $k=3\times6=18$ (2); polynomial $x^2-6x+18$ (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 6	Remainder $= p(1) = 6$
2	**(b)** 5	Other zero $= 9 - 4$
3	**(a)** 37	$(\alpha+\beta)^2 - 2\alpha\beta = 5^2 - 2(-6)$
4	**(a)** $\frac{8}{15}$	$\frac{\alpha+\beta}{\alpha\beta}$
5	**(c)** 20	$4^2 - 9(4) + k = 0$
6	**(c)** -2	$-2^2 + -2k - 8 = 0$
7	**(a)** 22	Remainder $= p(3) = 22$
8	**(a)** 5	Other zero $= 8 - 3$
9	**(c)** 65	$(\alpha+\beta)^2 - 2\alpha\beta = 9^2 - 2(8)$
10	**(b)** $\frac{2}{3}$	$\frac{\alpha+\beta}{\alpha\beta}$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$p(2) = 0$: $-14 + k = 0$ (1) so $k = 14$ (1).	1+1
12	New sum $= 2(5) = 10$, product $= 4(6) = 24$ (1); polynomial $x^2 - 10x + 24$ (1).	1+1
13	$\alpha+\beta = 3$, $\alpha\beta = -10$ (1); $\alpha^2+\beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = 29$ (1).	1+1
14	$p(4) = 0$: $-12 + k = 0$ (1) so $k = 12$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\alpha+\beta = 9$, $\alpha\beta = 18$ (1); $\frac{\alpha+\beta}{\alpha\beta} = \frac{1}{2}$ (1); $\alpha^2+\beta^2 = 45$ (1).	1+1+1
16	Sum of new zeroes $= \frac{\alpha+\beta}{\alpha\beta} = \frac{7}{12}$ (1); product $= \frac{1}{\alpha\beta} = \frac{1}{12}$ (1); polynomial $x^2 - (\frac{7}{12})x + (\frac{1}{12})$ (1).	1+1+1
17	$p(7) = 0$: $49 - 70 + 21 = 0$ gives $k = -10$ (1); sum of zeroes $= -k = 10$ (1); other zero $= 3$ (1).	1+1+1
	OR: $\alpha+\beta = 5$, $\alpha\beta = -14$ (1); $\frac{\alpha+\beta}{\alpha\beta} = -\frac{5}{14}$ (1); $\alpha^2+\beta^2 = 53$ (1).	1+1+1
18	Sum of new zeroes $= \frac{\alpha+\beta}{\alpha\beta} = \frac{7}{10}$ (1); product $= \frac{1}{\alpha\beta} = \frac{1}{10}$ (1); polynomial $x^2 - (\frac{7}{10})x + (\frac{1}{10})$ (1).	1+1+1
	OR: $p(-4) = 0$: $16 - 36 + 20 = 0$ gives $k = 9$ (1); sum of zeroes $= -k = -9$ (1); other zero $= -5$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Quadratic factor $(x-2)(x-3) = x^2 - 5x + 6$ (1); divide to get $x + 4$ (2); third zero -4 (1); zeroes $2, 3, -4$ (1).	1+2+1+1
	OR: Sum $= 7$, product $= k$ (1); $k = 2 \times 7 = 14$ (2); polynomial $x^2 - 7x + 14$ (2).	1+2+2
20	Quadratic factor $(x-1)(x+2) = x^2 + x - 2$ (1); divide to get $x - 6$ (2); third zero 6 (1); zeroes $1, -2, 6$ (1).	1+2+1+1
	OR: Sum $= 4$, product $= k$ (1); $k = 5 \times 4 = 20$ (2); polynomial $x^2 - 4x + 20$ (2).	1+2+2

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Polynomials** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 30	$5^2 - 11(5) + k = 0$
2	**(d)** -2	$5^2 + 5k - 15 = 0$
3	**(b)** 12	Remainder $= p(2) = 12$
4	**(d)** 7	Other zero $= 10 - 3$
5	**(a)** 34	$(\alpha+\beta)^2 - 2\alpha\beta = 2^2 - 2(-15)$
6	**(d)** $\frac{9}{20}$	$\frac{\alpha+\beta}{\alpha\beta}$
7	**(d)** 24	$6^2 - 10(6) + k = 0$
8	**(a)** -1	$-3^2 + -3k - 12 = 0$
9	**(b)** 15	Remainder $= p(2) = 15$
10	**(c)** 8	Other zero $= 11 - 3$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	New sum $= 2(3) = 6$, product $= 4(-4) = -16$ (1); polynomial $x^2 - 6x - 16$ (1).	1+1
12	$\alpha+\beta = 10$, $\alpha\beta = 9$ (1); $\alpha^2+\beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = 82$ (1).	1+1
13	$p(5) = 0$: $-5 + k = 0$ (1) so $k = 5$ (1).	1+1
14	New sum $= 2(8) = 16$, product $= 4(15) = 60$ (1); polynomial $x^2 - 16x + 60$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\alpha+\beta = 11$, $\alpha\beta = 18$ (1); $\frac{\alpha+\beta}{\alpha\beta} = \frac{11}{18}$ (1); $\alpha^2+\beta^2 = 85$ (1).	1+1+1
16	Sum of new zeroes $= \frac{\alpha+\beta}{\alpha\beta} = \frac{5}{12}$ (1); product $= \frac{1}{\alpha\beta} = \frac{1}{24}$ (1); polynomial $x^2 - (\frac{5}{12})x + (\frac{1}{24})$ (1).	1+1+1
17	$p(8) = 0$: $64 - 112 + 48 = 0$ gives $k = -14$ (1); sum of zeroes $= -k = 14$ (1); other zero $= 6$ (1).	1+1+1
	OR: $\alpha+\beta = 12$, $\alpha\beta = 32$ (1); $\frac{\alpha+\beta}{\alpha\beta} = \frac{3}{8}$ (1); $\alpha^2+\beta^2 = 80$ (1).	1+1+1
18	Sum of new zeroes $= \frac{\alpha+\beta}{\alpha\beta} = \frac{9}{8}$ (1); product $= \frac{1}{\alpha\beta} = \frac{1}{8}$ (1); polynomial $x^2 - (\frac{9}{8})x + (\frac{1}{8})$ (1).	1+1+1
	OR: $p(-7) = 0$: $49 - 28 - 21 = 0$ gives $k = 4$ (1); sum of zeroes $= -k = -4$ (1); other zero $= 3$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Quadratic factor $(x-3)(x-4) = x^2 - 7x + 12$ (1); divide to get $x + 1$ (2); third zero -1 (1); zeroes $3, 4, -1$ (1).	1+2+1+1
	OR: Sum $= 8$, product $= k$ (1); $k = 3 \times 8 = 24$ (2); polynomial $x^2 - 8x + 24$ (2).	1+2+2
20	Quadratic factor $(x-2)(x-5) = x^2 - 7x + 10$ (1); divide to get $x + 3$ (2); third zero -3 (1); zeroes $2, 5, -3$ (1).	1+2+1+1
	OR: Sum $= 9$, product $= k$ (1); $k = 2 \times 9 = 18$ (2); polynomial $x^2 - 9x + 18$ (2).	1+2+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.